

DATA 505: Homework 1

Variable assignment and basic arithmetic

2025-09-02

Context

In [Lecture 1](#), we talked about R, its history, its strengths and weaknesses, and looked at an example that highlights its usefulness for data analysis. In [Lab 1](#), we had our first use of Quarto to produce documents that include integrated R code.

In the next class, we will discuss some basics of R variables, types, and functions.

In between, with this first homework assignment, we will learn about: - Objects and variable assignment in R, and - Syntax for simple arithmetic with numeric variables in R. (“R as a calculator.”)

References

If the concept of variable assignment or its implementation in R are unfamiliar to you, I recommend section 2.1 and 2.2 here of free online book:

- Grolemund, G. (2014). Hands-On Programming with R. O’Reilly Media. Retrieved from <https://rstudio-education.github.io/hopr/>.

Coding tasks

Create an R script and use it to perform these tasks.

1. Define a variable `x` and assign it the value 5¹
2. Define another variable `y` and assign it the value 3.

¹In R, you do not need to declare a variable’s type before assigning a value. In languages like C and Java, you must specify the type, e.g., `int x = 5;`.

3. Add `x` and `y` together. What is the result?
4. Subtract `y` from `x`. What is the result?
5. Multiply `x` and `y`. What is the result?
6. Divide `x` by `y`. What is the result?
7. Calculate `x` raised to the power of `y` (that is, x^y). What is the result?
8. Use the `sqrt()` function to find the square root of `x`.
9. Assign the value -10 to a variable called `z`. Use the `abs()` function to find the absolute value of `z`.
10. Try dividing `x` by 0. What happens? What result does R give you?
11. Assign a new value to `x` (for example, 12). What happens to the old value?
12. Use the `#` symbol to add a comment to your script explaining what one of your lines of code does.
13. (Challenge) Combine arithmetic and a function: Calculate the square root of the sum of `x` squared and `y` squared.

Side note: Variable assignment syntax

A few different syntactical styles are valid for assigning values to variables in R.

```
x <- 4 # most common in R style convention
x = 4 # less common in R, but still works
4 -> x # awkward but works
x <- y <- 4 # works: assigns to both x and y
x <- 4 -> y # also works but very eccentric (discourage doing this)
```

Others are invalid, like:

```
x == 4 # this is logical comparison, not assignment
x <= 4 # this is an inequality, not assignment
```